Volumes of Revolution

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.02

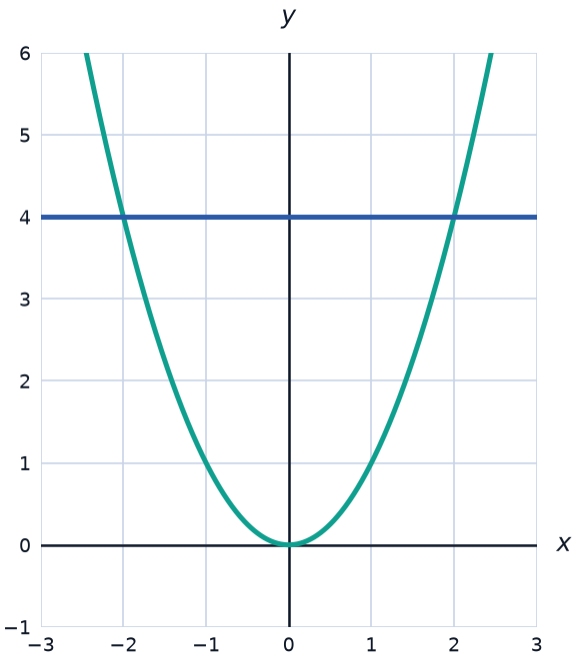
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: The disk and washer methods

Find the volume generated when the region under $y = \sqrt{x}$ on $[0, 4]$ is revolved about the x-axis.

2 [Foundational] • Target: The disk and washer methods

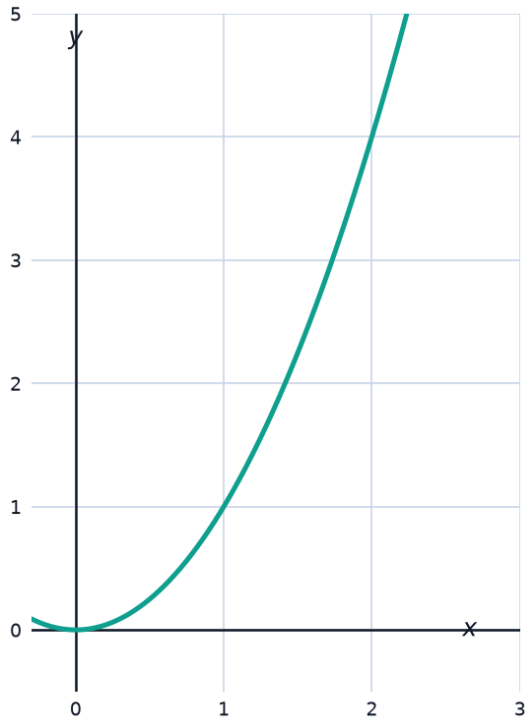
The region between $y = x^2$ and $y = 4$ is revolved about the x-axis. Using the washer method, set up and evaluate the volume integral.



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3 [Foundational] • Target: The shell method

The region under $y = x^2$ on $[0, 2]$ is revolved about the y-axis. Use the shell method to find the volume.



4 [Foundational] • Target: The shell method

The region bounded by $y = x$, $y = 0$, and $x = 3$ is revolved about the y-axis. Find the volume using the shell method.

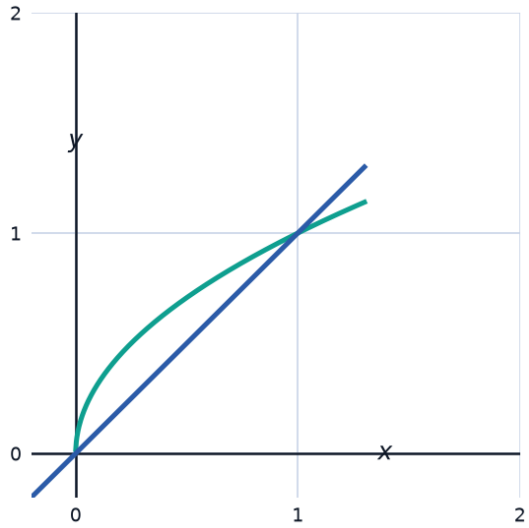
5 [Foundational] • Target: Comparing methods

The region under $y = x^2$ on $[0, 2]$ is revolved about the x-axis (not the y-axis). Find the volume using the disk method.

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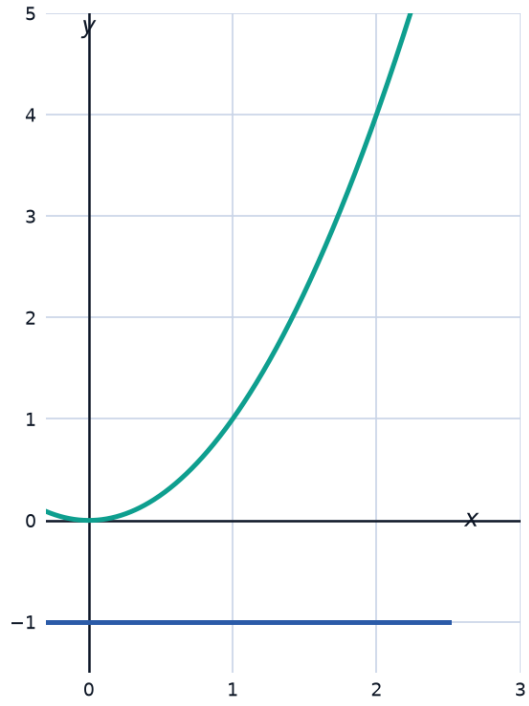
6 [Foundational] • Target: Comparing methods

The region bounded by $y = \sqrt{x}$ and $y = x$ (between their two intersection points) is revolved about the x-axis. Find the volume using the washer method.



7 [Foundational] • Target: Volumes about other axes

The region under $y = x^2$ on $[0, 2]$ is revolved about the line $y = -1$. Find the volume, using outer radius $x^2 + 1$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

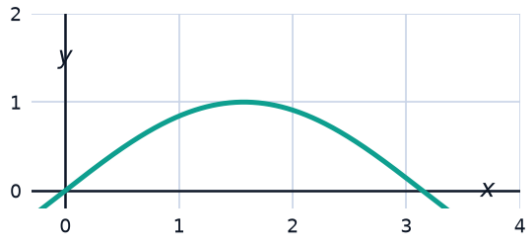
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8 [Intermediate] • Target: Volumes about other axes

The region under $y = x^2$ on $[0, 2]$ is revolved about the line $x = 3$ (using the shell method with radius $3 - x$).

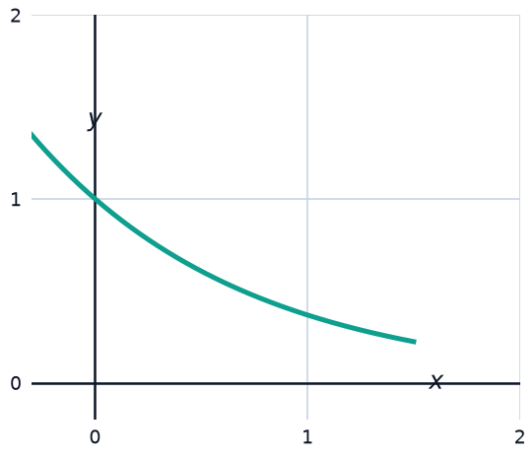
9 [Intermediate] • Target: Volumes with trigonometric and exponential curves

The region under $y = \sin x$ on $[0, \pi]$ is revolved about the x-axis. Find the volume, using $\sin^2 x = \frac{1 - \cos 2x}{2}$.



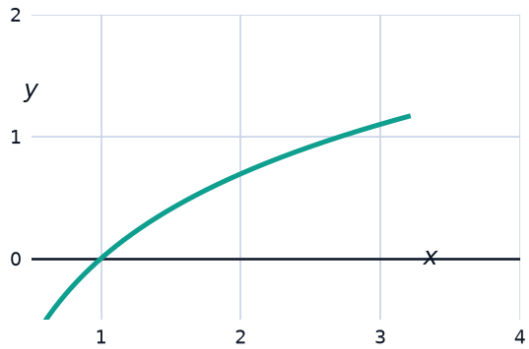
10 [Intermediate] • Target: Volumes with trigonometric and exponential curves

The region under $y = e^{-x}$ on $[0, 1]$ is revolved about the x-axis. Find the volume, to three decimal places.



11 [Intermediate] • Target: Setting up without evaluating

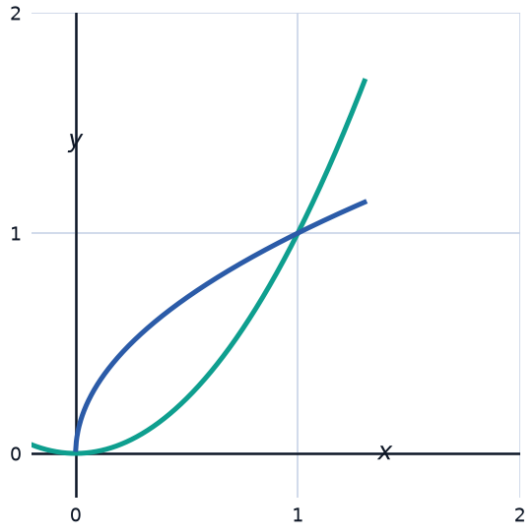
Set up (do not evaluate) the volume integral for the region under $y = \ln x$ on $[1, e]$, revolved about the x-axis.



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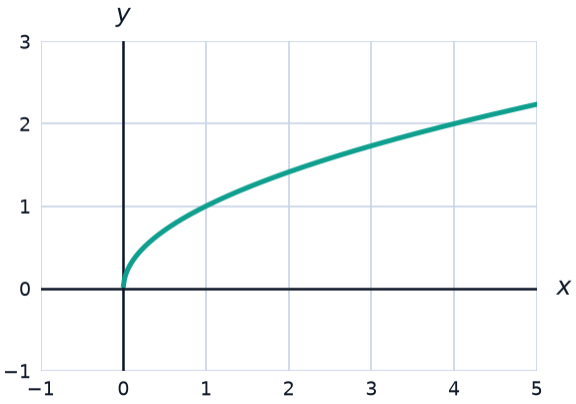
12 **[Intermediate]** • **Target: Setting up without evaluating**

Set up (do not evaluate) the volume integral for the region between $y = x^2$ and $y = \sqrt{x}$ (on $[0, 1]$) revolved about the y-axis, using the shell method.



13 **[Intermediate]** • **Target: Visualizing the generating region**

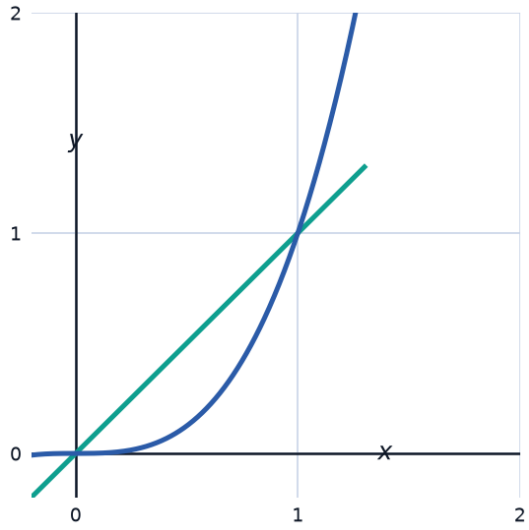
The region under $y = \sqrt{x}$ on $[0, 4]$ (from an earlier disk-method problem) is shown. Sketching this region mentally revolved about the x-axis produces a solid — describe its shape, and confirm the volume 8π found earlier is reasonable by comparing to a cylinder of radius 2 and length 4.



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14 **[Intermediate]** • **Target: Volume of a solid with a hole (washer method practice)**

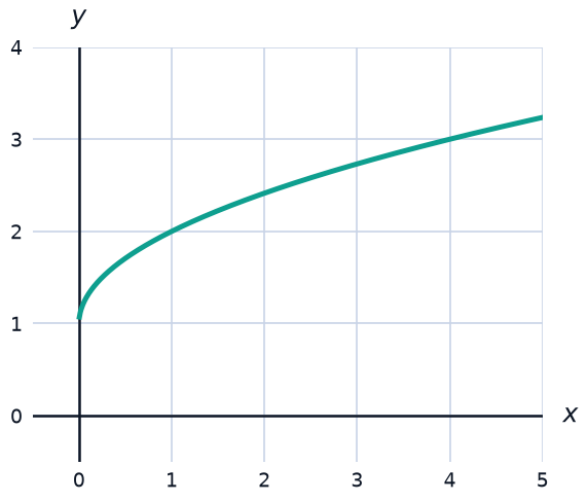
The region between $y = x$ and $y = x^3$ on $[0, 1]$ is revolved about the x-axis. Find the volume using the washer method (outer radius x, inner radius x^3).



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: An applied volume problem** • **Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain**

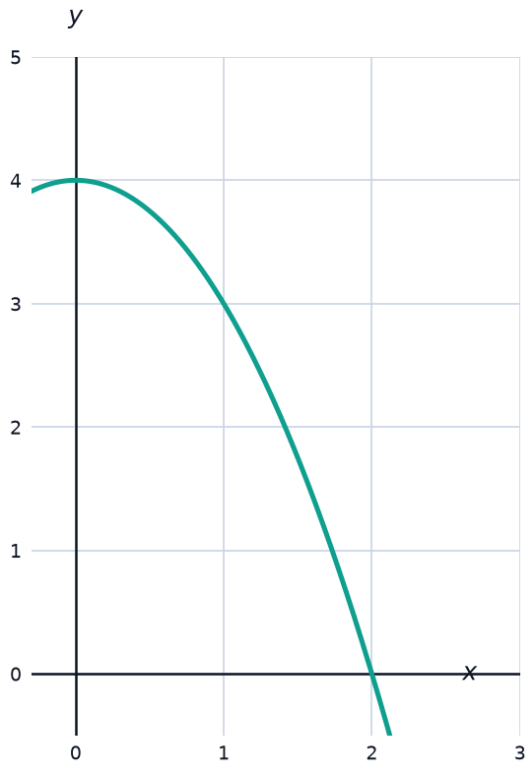
A vase is modeled by revolving the region under $y = \sqrt{x} + 1$ on $[0, 4]$ about the x-axis, where x is the height (cm) and y is the radius. Find the volume of the vase.



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16 [Challenging] • Target: Volume with the shell method, revisited • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

The region bounded by $y = 4 - x^2$ and $y = 0$ (for $x \geq 0$) is revolved about the y-axis. Use the shell method to find the volume.



17 [Challenging] • Target: Comparing methods • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Shell method for solid of revolution about y-axis: cylindrical shells have radius $=$.

18 [Challenging] • Target: Volumes about other axes • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Volume of solid of revolution when region $0 \leq y \leq e^{-x^2}$, $0 \leq x \leq 1$ is rotated about y-axis (shell):.

19 [Challenging] • Target: Volumes with trigonometric and exponential curves • Cross-Lesson: 6.01 Related Rates

Volume by shell about x-axis: shells have radius $=$.

20 [Challenging] • Target: Setting up without evaluating • Cross-Lesson: 6.02 Optimization and Curve Sketching

Volume of solid of revolution of $y = \cos x$ about x-axis from 0 to $\pi/2$:.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Disk Method (Rotation Around Axes)

Find the volume of the solid generated by revolving $y = \sqrt{x}$ from $x = 0$ to $x = 4$ about the x-axis.

a. 8π

b. 16π

c. 4π

d. $32\pi/3$

22 [Foundational] • Target: Disk Method (Rotation Around Axes)

Find the volume when the region under $y = x^2$ from $x = 0$ to $x = 2$ is rotated about the x-axis.

a. $\frac{32\pi}{5}$

b. $\frac{16\pi}{5}$

c. 8π

d. $\frac{64\pi}{5}$

23 [Foundational] • Target: Disk Method (Rotation Around Axes)

Find the volume when the region between $x = \sqrt{y}$, $y = 0$, and $y = 4$ is revolved about the y-axis.

a. 8π

b. 16π

c. 4π

d. $32\pi/3$

24 [Foundational] • Target: Washer Method (Hollow Solids)

Find the volume when the region bounded by $y = x$ and $y = x^2$ is revolved about the x-axis.

a. $\frac{2\pi}{15}$

b. $\frac{\pi}{15}$

c. $\frac{4\pi}{15}$

d. $\frac{\pi}{5}$

25 [Foundational] • Target: Washer Method (Hollow Solids)

The region bounded by $y = 4 - x^2$ and $y = 0$ is rotated about the line $y = -1$. Set up the outer and inner radii.

a. $R(x) = (4 - x^2) - (-1) = 5 - x^2$, $r(x) = 0 - (-1) = 1$

b. $R(x) = 4 - x^2$, $r(x) = 1$

c. $R(x) = 5 - x^2$, $r(x) = 0$

d. $R(x) = 4 - x^2 + 1$, $r(x) = 2$

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26 [Foundational] • Target: Washer Method (Hollow Solids)

Evaluate the volume in the previous problem rotated about $y = -1$ on $[-2, 2]$.

a. $\frac{384\pi}{15} = \frac{128\pi}{5}$

b. $\frac{64\pi}{5}$

c. 32π

d. $\frac{256\pi}{5}$

27 [Foundational] • Target: Cylindrical Shells Method

State the cylindrical shells formula for a region under $y = f(x)$ on $[a, b]$ revolved about the y-axis.

a. $V = 2\pi\int_a^b xf(x)\,dx$

b. $V = \pi\int_a^b [f(x)]^2\,dx$

c. $V = 2\pi\int_a^b [f(x)]^2\,dx$

d. $V = \pi\int_a^b x^2 f(x)\,dx$

28 [Foundational] • Target: Cylindrical Shells Method

Find the volume when the region under $y = 3x - x^2$ from $x = 0$ to $x = 3$ is revolved about the y-axis.

a. $\frac{27\pi}{2}$

b. 27π

c. $\frac{81\pi}{4}$

d. $\frac{9\pi}{2}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Cylindrical Shells Method

Use cylindrical shells to find the volume when $y = e^{-x^2}$ on $[0, 1]$ is revolved about the y-axis.

a. $\pi(1 - e^{-1})$

b. $2\pi(1 - e^{-1})$

c. πe^{-1}

d. $\frac{\pi}{2}(1 - e^{-1})$

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30 [Intermediate] • Target: Cross-Sections with Known Geometric Shapes

The base of a solid is the circle $x^2 + y^2 = 4$. Cross-sections perpendicular to the x-axis are squares. Find the volume.

a. $\frac{128}{3}$

b. 16π

c. 32

d. $\frac{64}{3}$

31 [Intermediate] • Target: Cross-Sections with Known Geometric Shapes

The base of a solid is bounded by $y = x^2$ and $y = 4$. Cross-sections perpendicular to the y-axis are semicircles. Find the volume.

a. 4π

b. 2π

c. 8π

d. $\frac{8\pi}{3}$

32 [Intermediate] • Target: Cross-Sections with Known Geometric Shapes

The base is the triangle with vertices $(0, 0), (1, 0), (0, 1)$. Cross-sections perpendicular to the x-axis are equilateral triangles. Find the volume.

a. $\frac{\sqrt{3}}{12}$

b. $\frac{\sqrt{3}}{6}$

c. $\frac{1}{6}$

d. $\frac{\sqrt{3}}{4}$

33 [Intermediate] • Target: Cylindrical Shells Method

Volumes: for the same region rotated, which is larger: shell or washer?

a. Shell is always larger

b. They give the same volume

c. Washer is always larger

d. Depends on axis and region

34 [Intermediate] • Target: Cross-Sections with Known Geometric Shapes

Volume of solid with square cross-sections perpendicular to x-axis, bounded by $y = \sqrt{x}$ and $x = 4$:

a. 4

b. 8

c. 16/3

d. 8/3

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35 [Intermediate] • Target: The disk and washer methods

Volume of solid with circular cross-sections of radius $f(x)$:

a. $\int_a^b \pi f(x)dx$

b. $\int_a^b f(x)^2 dx$

c. $\int_a^b \pi[f(x)]^2 dx$

d. $2\pi\int_a^b f(x)dx$

36 [Intermediate] • Target: The shell method

Which gives volume by the disk method about the y-axis?

a. $\int_a^b f(y)\,dy$

b. $\pi\int_a^b [f(x)]^2\,dx$

c. $2\pi\int_a^b xf(x)\,dx$

d. $\pi\int_c^d [f(y)]^2\,dy$ (solve for x first)

37 [Intermediate] • Target: Comparing methods

Volume of the solid of revolution of $y = \sqrt{9 - x^2}$ (semicircle) about x-axis:

a. 18π

b. $\frac{4}{3}\pi \cdot 27 = 36\pi$

c. 36π

d. $\frac{4}{3}\pi(3)^3 = 36\pi$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Volumes about other axes • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Volume of revolution of $y = x^3$ about x-axis from 0 to 1:

a. π

b. $\pi/5$

c. $\pi/7$

d. $3\pi/7$

39 [Challenging] • Target: Volumes with trigonometric and exponential curves • Cross-Lesson: 6.01 Related Rates

The region under $y = \sin(x)$ for $0 \le x \le \pi$ is revolved about the x-axis. Find the volume.

a. $\frac{\pi}{2}$

b. π^2

c. $\frac{\pi^2}{2}$

d. $2\pi^2$

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40 [Challenging] • Target: Setting up without evaluating • Cross-Lesson: 6.02 Optimization and Curve Sketching

The region under $y = e^x$ for $0 \le x \le \ln(2)$ is revolved about the x-axis. Find the volume.

a. $\frac{3}{2}$

b. 3π

c. $\frac{3\pi}{2}$

d. $2\pi(-1 + \ln(4))$

41 [Challenging] • Target: Visualizing the generating region • Cross-Lesson: 7.01 Integration by Substitution

The region under $y = e^x$ for $0 \le x \le \ln(5)$ is revolved about the x-axis. Find the volume.

a. 12

b. 12π

c. 24π

d. $2\pi(-4 + \ln(3125))$

42 [Challenging] • Target: Volume of a solid with a hole (washer method practice) • Cross-Lesson: 7.02 Integration by Parts

The region under $y = \sqrt{x}$ for $0 \le x \le 3$ is revolved about the x-axis. Find the volume.

a. $\frac{9}{2}$

b. $\frac{9\pi}{2}$

c. 9π

d. $\frac{36\sqrt{3}\pi}{5}$

43 [Challenging] • Target: An applied volume problem • Cross-Lesson: 8.01 Area Between Curves

The solid whose cross-sections perpendicular to the x-axis are equilateral triangles with side $f(x)$ has:

a. $V = \frac{\sqrt{3}}{4}\int f^2\,dx$

b. $V = \int f\,dx$

c. $V = \pi\int f^2\,dx$

d. $V = 3\int f\,dx$

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44 [Challenging] • Target: Volume with the shell method, revisited • Cross-Lesson: 8.03 Arc Length and Improper Integrals

The region under $y = \sqrt{x}$ for $0 \le x \le 4$ is revolved about the x-axis. Find the volume.

a. 8

b. 8π

c. 16π

d. $\frac{128\pi}{5}$

45 [Challenging] • Target: Disk Method (Rotation Around Axes) • Cross-Lesson: 9.01 Separable Differential Equations

Volume by shells revolving $y = x^2$ about the y-axis from 0 to 1:

a. $\pi/2$

b. $\pi/4$

c. π

d. $2\pi/4$

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